

MEDINEXUS AI

Intelligent Healthcare Management System

Disease Prediction • Medical Report Analysis • Real-Time Hospital Management

MediNexus AI is a full-stack healthcare platform that connects patients, doctors, and hospital administrators through AI-assisted disease prediction, centralized patient management, secure role-based access, REST APIs, and real-time WebSocket synchronization.

Developer

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01 | Project Overview

MediNexus AI is an AI-powered healthcare management platform connecting **Patients, Doctors, and Hospital Administrators** through a centralized real-time ecosystem. It combines AI-assisted disease prediction and medical report analysis with role-based access, REST APIs, centralized data management, notifications, and WebSocket synchronization.

The centralized database acts as the single source of truth for patients, predictions, doctor assignments, and notifications.

Core Modules

- **Patient Portal:** registration, authentication, profile, symptoms, disease prediction, reports, history and doctor communication.
- **Doctor Portal:** authentication, assigned patients, clinical information, prescriptions, consultation history and notifications.
- **Hospital Admin Portal:** Patient Log, doctor onboarding, assignments, risk monitoring, notifications and hospital operations.

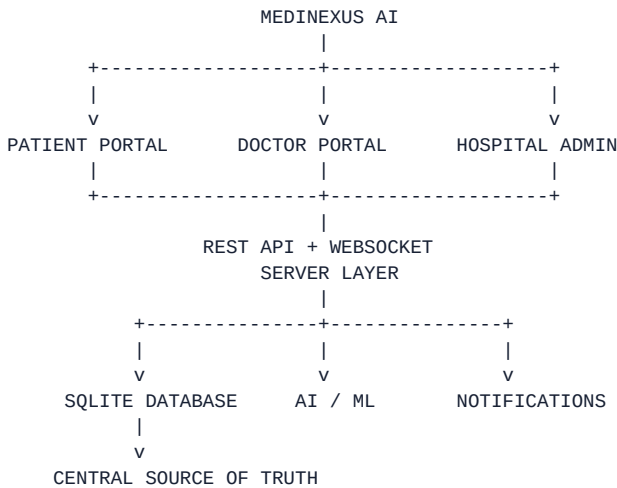
Objectives

- Build a centralized healthcare management ecosystem.
- Assist with early disease-risk assessment using AI/ML.
- Improve patient-doctor communication and information flow.
- Reduce manual synchronization between hospital portals.
- Provide real-time updates across authorized portals.

02 | Feature Matrix

Module	Key Capabilities
Patient	Registration, authentication, profile, symptoms, prediction, reports, history
Doctor	Authentication, assigned patients, clinical records, prescriptions, consultation
Admin	Patient Log, doctor management, pre-authorization, assignments, operations
AI/ML	Symptom-based prediction, risk evaluation, medical information analysis
Real-Time	WebSocket events, cache invalidation, dashboard updates
Notifications	Registration, prediction and doctor-assignment alerts

03 | System Architecture



Real-Time Event Flow

PATIENT_REGISTERED -> Database -> Admin Notification -> Broadcast
DISEASE_PREDICTED -> Prediction -> Risk Metrics -> Broadcast
DOCTOR_ASSIGNED -> Patient + Doctor Notifications -> Broadcast

04 | Database & Data Integrity

The centralized SQLite database maintains consistent healthcare data across all portals. Patient-to-doctor assignment is stored at database level so every portal reads the same authoritative mapping.

Entity	Purpose
Patients	Patient identity, profile, contact and assigned doctor
Doctors	Doctor identity, specialty, authorization and access
Predictions	Prediction history and risk information
Notifications	Persistent role-specific notifications

```
Patients
|- Assigned_Doctor_ID
|- Assigned_Doctor_Name
|
+-- Patients <-> Predictions
+-- Patients <-> Doctors
+-- Roles    <-> Notifications
```

05 | REST API Reference

Method	Endpoint	Purpose
POST	/api/patient/register	Register patient + admin notification
GET	/api/patients	Secure patient directory
GET	/api/patient/:id	Patient clinical data
POST	/api/prediction	AI disease evaluation
GET	/api/prediction/:id	Prediction history
GET	/api/hospital/patient-log	Admin patient log
POST	/api/assign-doctor	Assign/change primary doctor
GET	/api/notifications	Role-specific notifications
PUT	/api/patient/update	Update patient profile
GET	/api/doctor/patients	Doctor-specific patients

06 | Security & Access Control

Role	Access Scope
Patient	Own profile, symptoms, predictions, reports and history
Doctor	Authorized clinical information and assigned patients
Admin	Hospital-wide patient, doctor, notification and operational management

Doctor onboarding is controlled through the Hospital Admin Portal. Pre-authorized doctors can be provisioned with hospital credentials before accessing the Doctor Portal.

GitHub Security: Never commit real patient records, passwords, API keys, .env files, production credentials, or databases containing personal healthcare information.

07 | Technology Stack

Layer	Technologies
Frontend	React.js, JavaScript, HTML, CSS
Backend	Node.js, REST APIs, WebSockets
Database	SQLite
AI / ML	Python, Machine Learning, Disease Prediction
Visualization	Chart.js, dashboard components
Tools	Git, GitHub, VS Code, Linux, AWS basics

08 | Hospital Admin Patient Log

The Comprehensive Patient Log centralizes patient information for hospital administrators.

- Patient ID, patient name, age and gender.
- Email, phone and registration date.
- Latest disease prediction and dynamic risk level.
- Interactive doctor assignment.
- Triage/history actions and clinical management.

09 | Notification & Event System

Event	System Action
Patient Registration	Persistent admin notification + PATIENT_REGISTERED broadcast
Disease Prediction	Prediction notification + DISEASE_PREDICTED broadcast
Doctor Assignment	Patient and doctor notifications + DOCTOR_ASSIGNED broadcast
Profile Update	Synchronizes updated information across authorized clients

10 | End-to-End Workflow

```
graph TD
    A[Patient Registration] --> B[Centralized Database]
    B --> C[Symptom / Report Input]
```

```

|
AI / ML Prediction
|
Risk Evaluation
|
Hospital Admin Review
|
Doctor Assignment
|
Doctor Portal
|
Clinical Consultation
|
Notifications + Real-Time Synchronization

```

11 | GitHub Repository Setup

Recommended Repository

MediNexus-AI

Repository Description

AI-powered healthcare management system integrating Patient, Doctor & Hospital Admin portals with disease prediction, medical report analysis, and real-time WebSocket synchronization.

Recommended Topics

artificial-intelligence machine-learning healthcare
healthcare-management disease-prediction medical-ai
python sqlite rest-api websocket react nodejs full-stack

Professional Repository Structure

```

MediNexus-AI/
|
+-- frontend/
|   +-- patient-portal/
|   +-- doctor-portal/
|   +-- admin-portal/
|
+-- backend/
|   +-- server/
|   +-- routes/
|   +-- controllers/
|   +-- database/
|   +-- websocket/
|
+-- ai-model/
|   +-- dataset/
|   +-- training/
|   +-- prediction/
|
+-- screenshots/
|   +-- patient-portal.png
|   +-- doctor-portal.png
|   +-- admin-portal.png
|
+-- README.md
+-- requirements.txt
+-- package.json
+-- .gitignore

```

12 | README.md — Ready-to-Use Content

Use the following as the core README content. Add screenshots, architecture images, and a demo link near the top of the live repository.

```
# ■ MediNexus AI

## Intelligent Healthcare Management System with Disease
## Prediction & Medical Report Analysis

MediNexus AI is an AI-powered healthcare management platform
connecting Patients, Doctors and Hospital Administrators through
a centralized real-time healthcare ecosystem.

### ■ Key Features
- Patient Portal
- Doctor Portal
- Hospital Admin Portal
- AI-based disease prediction
- Medical report analysis
- Centralized SQLite database
- REST APIs
- WebSocket real-time synchronization
- Role-based authentication
- Doctor assignment and notifications

### ■ Technology Stack
React.js | Node.js | Python | SQLite | REST API |
WebSockets | AI/ML | Git/GitHub
```

13 | Professional GitHub Presentation

For a strong GitHub presence, pin MediNexus AI near the top of your profile and keep the presentation consistent.

Profile Element	Recommended Content
Pinned Repository	MediNexus-AI
Description	One-line AI + healthcare + real-time synchronization summary
README Hero	Project title + short tagline + technology badges
Screenshots	Patient, Doctor and Admin portal screenshots
Architecture	System architecture diagram
Demo	Deployed application/demo link, if public
Documentation	Setup, API, workflow, database and security sections

14 | Professional README Badges

```
![Python](https://img.shields.io/badge/Python-3.x-blue)
![React](https://img.shields.io/badge/React.js-Frontend-61DAFB)
![Node.js](https://img.shields.io/badge/Node.js-Backend-green)
![SQLite](https://img.shields.io/badge/SQLite-Database-003B57)
![WebSocket](https://img.shields.io/badge/WebSocket-Real--Time-orange)
![AI/ML](https://img.shields.io/badge/AI%2FML-Disease%20Prediction-purple)
```

15 | Future Scope

- Electronic Health Record (EHR) integration.
- Advanced OCR for handwritten and scanned medical reports.
- Improved disease prediction models and evaluation.
- Wearable health-device integration.
- Cloud deployment and scalable multi-hospital architecture.
- Advanced analytics and hospital intelligence dashboards.

- Appointment and bed-availability optimization.
- Multi-hospital and enterprise role-based support.

16 | 30-Second Project Pitch

MediNexus AI is a full-stack intelligent healthcare management system that connects patients, doctors, and hospital administrators through a centralized platform. It combines AI-based disease prediction and medical report analysis with secure role-based portals, REST APIs, centralized SQLite data, and WebSocket-based real-time synchronization. The system synchronizes patient registrations, predictions, doctor assignments, and notifications across the healthcare ecosystem.

17 | Final GitHub Checklist

- Create repository: **MediNexus-AI**.
- Add a professional repository description and relevant topics.
- Upload a clean source-code structure.
- Add the README.md content from this document.
- Add screenshots of Patient, Doctor and Admin portals.
- Add a system architecture diagram.
- Add a live demo link only if it is safe and publicly accessible.
- Create a strong .gitignore and remove secrets before pushing.
- Do not upload real patient/medical data or credentials.
- Pin the repository on your GitHub profile.